

Institutional Positioning from 13F Filings: Point-in-Time Data Engineering, a Signal Taxonomy, and What Survived a Pre-Registered Research Funnel

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Abstract

We build a point-in-time research system for SEC 13F institutional holdings — bitemporal storage keyed on filing acceptance, amendment semantics from cover pages, a CUSIP→name→current-ticker crosswalk with human-in-the-loop resolution — and run a pre-registered research funnel over six classes of positioning signals. Two objects survive. The first, *New-Conviction Breadth* (NCB): the cross-section of filers opening or materially growing top-decile-of-book positions, residualized on size and liquidity, earns +1.9% per quarter in quintile spreads ($t = +3.4$, Sharpe ≈ 1.0), replicates eight times across two universes and four implementations, and its full 36-point hyperparameter surface is positive out-of-sample. The second, *Forced-Sale Supply* (FSS): managers whose implied flow falls below -10% sell 34.5% of held shares the following quarter against 23.1% for healthy managers ($t \approx +10$ at every threshold), and the resulting supply basket prices as a regime-dependent short. Around these survivors we document, with the same protocol, why almost everything else fails: fifteen crowding formulations, nine smart-money designs, five pre-registered interaction effects, and daily anticipation of institutional stress — one instance of which we retract explicitly after a data-coverage audit, and the audit itself (46% of book value initially unmapped by name matching) is presented as a core result. A Fama–MacBeth combiner over ten characteristics is the production model (OOS Sharpe 0.86); IPCA and gradient-boosted alternatives are evaluated and held at candidate status by the funnel’s own promotion rules. Two empirical regularities organize every outcome: ownership *states* carry no directional information at any clock, and alpha lives only where the observed action was *costly* (conviction displacement) or *forced* (redemption-driven liquidation).

Contents

1	Introduction	3
2	Data engineering: from EDGAR to a point-in-time store	4
2.1	Ingestion and storage	4
2.2	Amendments, value scale, identity	4
2.3	When information actually arrives	4
2.4	The crosswalk, and the audit that mattered	5
2.5	The re-verification campaign and one retraction	6
3	The manager universe: a census, not a chosen list	6

4	Evaluation protocol and research discipline	7
5	A taxonomy of positioning signals	8
5.1	The seventeen-signal catalogue	9
5.2	Replications, stated as such	9
6	New-Conviction Breadth (NCB)	10
6.1	Mechanism	10
6.2	Results and replications	11
6.3	Hyperparameters: an a-priori spec and a measured surface	12
6.4	A machine-learning cross-check that found the same thing	12
7	Forced-Sale Supply (FSS)	12
7.1	The flow instrument	12
7.2	Threshold justification without selection	13
7.3	From managers to stocks, and what the price says	13
7.4	Prediction fails at every clock (Law 2)	14
8	Combining models and net-of-cost production results	14
8.1	The Fama–MacBeth production combiner	14
8.2	The IPCA challenger: is the premium a latent-factor loading?	15
8.3	The net engine	15
8.4	Long-only implementation	16
9	Latent structure: ICA on flows, NMF on portfolios	18
9.1	Why ICA and not PCA, from the data up	18
9.2	NMF latent strategies	20
10	Event-clock studies, monitors, and tactical results	20
11	Context: why imitating these books is a losing design	22
12	The graveyard, and the two laws it obeys	22
13	Limitations and next steps	23
A	Signal definitions	26
B	Module compendium	26
C	Hyperparameter surfaces	29

1 Introduction

Quarterly 13F filings are the most complete public record of institutional equity positioning, and the most heavily mined. The literature’s box score is mixed by construction: the filings are stale (up to 45 days), long-only, and quarterly, so most of what they reveal is already in prices. Our starting premise is therefore not “institutions know something,” but a narrower mechanical claim: *the filings occasionally reveal actions that were either costly to take or impossible to avoid*, and only those actions should carry a premium. A manager displacing an existing top-decile position to make room for a new name has paid a real price for that opinion; a manager liquidating a third of a book under redemptions is not expressing an opinion at all. Everything in this report is organized around whether a signal isolates one of those two situations.

The work divides into three parts. The first (Sections 2–3) is data engineering, and we give it the space usually reserved for results because in this dataset the engineering *is* the result: a bitemporal point-in-time store keyed on filing acceptance timestamps; amendment semantics read from cover pages rather than guessed; per-filing value scale detection; a CUSIP→issuer-name→current-ticker crosswalk with human resolution of edge cases; and a value-weighted coverage audit that late in the project found 46.4% of book value unmapped — EDGAR’s name abbreviations defeat exact matching precisely on the mega-caps — forcing a fix and a 24-run re-verification campaign in which one previously reported result was retracted (Section 2.5). We report the retraction with its mechanism because the episode is the strongest evidence we can offer that the rest of the numbers mean something.

The second part (Sections 5–7) is a taxonomy: six classes of positioning signals, seventeen catalogue constructions evaluated under one fixed protocol, five pre-registered interaction effects, and the two survivors. New-Conviction Breadth (NCB) counts, per stock, the fraction of filers that opened or grew ($\geq 25\%$, split-adjusted) a position sitting in the top decile of their own book, residualized on size and dollar liquidity because a formal placebo (“dart-throwing value-weighted managers”) shows the raw count is mechanically correlated with both. Forced-Sale Supply (FSS) converts a validated manager-level flow instrument into a stock-level supply measure. For each we present the economic mechanism first, the returns second, and the hyperparameter surface third — every parameter in this report is either justified by a measurement or explicitly labeled a convention.

The third part (Sections 8–13) assembles the pieces into production objects and stress-tests the assembly: a Fama–MacBeth expected-return combiner over ten characteristics (the production model), an IPCA challenger under the Kelly, Pruitt and Su [2019] instrumented-loadings framework, a gradient-boosted ex-post exercise that rediscovers NCB and finds no nonlinear premium, net-of-cost engine results with explicit commission and borrow accounting, and the descriptive context — the median large 13F filer underperforms the S&P 500 by 3.3% per year before fees — that explains why imitating these books is a losing design and harvesting their *flows* is not.

Two regularities recur so often that we state them as laws and tag every result with them. **Law 1: states have no arrow.** Levels of ownership, breadth, crowding, concentration, or factor tilt never predict returns in this data, at any clock, in fifteen independent formulations; changes and constrained actions do. **Law 2: distress is information when observed, never when predicted.** Contemporaneous forced selling is measurable with $t \approx +10$; every attempt to anticipate it — daily synthetic stress, quarterly flow forecasting, single-name crash cascades — fails its mechanism gate or dies on the full panel.

2 Data engineering: from EDGAR to a point-in-time store

2.1 Ingestion and storage

Two ingestion paths run in parallel by design. The SEC DERA structured datasets (~53 quarterly archives) are cheap and carry the other-manager graph, but begin in 2013Q2 and inherit the SEC’s own parsing unaudited. The raw-filing path (SGML envelopes; XML after 2013, fixed-width layouts before) covers the whole archive and is auditable, at one HTTP request per filing. A cross-validation routine re-parses a random sample of raw filings and compares row counts and value totals against DERA; neither source can check itself, and agreement below 95% halts the pipeline. The sample begins in 2013Q2 where the two paths overlap and the structured OTHERMANAGER graph is available — earlier eras are parseable but single-sourced, and we prefer a shorter sample with a second witness.

Every holding row carries two dates: the period it describes and the timestamp at which it became public (EDGAR acceptance). The store’s only read path for research, `as_of(period, knowledge_date)`, materializes both the current and prior quarter at the same knowledge date, so a quarter-over-quarter change never mixes what was known then with what is known now. Three mechanical look-ahead channels are closed at this layer: late filings (a filer’s book enters the panel only after its acceptance timestamp), amendments (a correction filed in March cannot retroactively improve a January decision), and restatement semantics.

2.2 Amendments, value scale, identity

Amendment handling follows the cover page, never a heuristic: **RESTATEMENT** replaces the filer’s table for the period; **NEW HOLDINGS** is additive (typically a lapsed confidential-treatment order); a missing declaration is treated as a restatement, the conservative choice. Misreading an additive amendment as a restatement silently deletes a quarter’s book — this failure mode is pinned by a test. An amendment is a *version* of its original, and never competes with it in any largest-book logic. Value scale (the 2022–2023 thousands-to-dollars cutover) is detected per filing keyed on the *filing* date, with loud outlier flags for the tail of filers that ignores the rule. Filer identity is resolved by union-find over explicit evidence (shared filings, CIK succession); name-similarity methods only propose, and any merge requires a manual entry in a reviewed configuration file — nothing in the pipeline merges on string distance alone.

2.3 When information actually arrives

The decision clock used throughout is quarter-end +45 calendar days, the statutory deadline. Table 1 and Figure 1 justify it with the acceptance-timestamp distribution over 254,630 original filings: the median filer reports 41 days after quarter end, half of all filings arrive in the final five days before the deadline, 89.4% have arrived by D+45 and 98.1% by D+60; the 99th percentile is 128 days and the maximum observed 400. Lateness is sticky (a filer late once is late again 41% of the time), so late books are a persistent subpopulation, not noise. A per-name ragged-entry design — entering each stock on the day its holder’s filing actually arrived — adds +0.2%/yr ($t \approx 0$) over the batch clock (Section 10): the information is so concentrated at the deadline that waiting for it costs nothing.

Table 1: Filing-delay distribution, original 13F filings 2013–2026 ($n = 254,630$). The D+45 batch clock is justified by measurement: 89% of books are public by the statutory deadline and the ragged per-name alternative adds nothing (Section 10).

	median	p90	p99	max	$\leq D+40$	$\leq D+45$	$\leq D+60$
delay (days)	41	46	128	400	49.9%	89.4%	98.1%

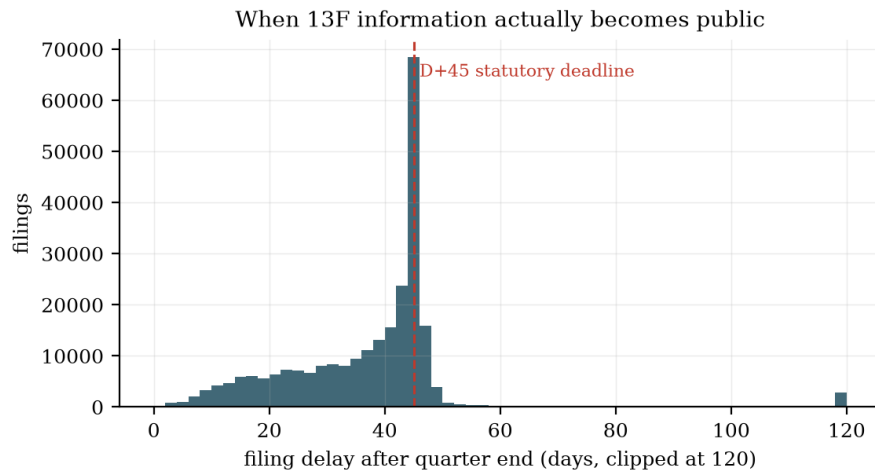


Figure 1: Acceptance-timestamp distribution. Half of all filings land in the last five days before the statutory deadline (dashed line).

2.4 The crosswalk, and the audit that mattered

Holdings identify securities by CUSIP; the price panel is keyed by ticker. The map is CUSIP $\rightarrow$ issuer name (from the filings themselves) $\rightarrow$ *current* ticker by normalized-name match against the exchange directory, ambiguous names discarded, with manual resolution of edge cases (human in the loop). Three failure modes were analyzed separately. Ticker renames (FB $\rightarrow$ META) are correct by construction, because the vendor reindexes history under the current symbol. Ticker recycling fails *closed*: history of a delisted issuer is never spliced under a recycled symbol, so affected periods drop out at the price floor — exclusion, never mispricing (4.5% of tickers flagged; inspected cases are known spin-offs and IPOs). CUSIP reassignment by the same issuer — the genuine blind spot — was measured directly: 1,574 events over 51 quarters, contaminating 0.22% of position-exit flow; exit-based signals are safe uncorrected.

Late in the project a value-weighted audit — prompted by the simple question “what happens when tickers change?” — found that **46.4% of 2025Q4 book value was unmapped**. The cause was not renames but EDGAR’s name abbreviations (“LILLY ELI & CO”, “INTERNATIONAL BUSINESS MACHS”), which defeat exact matching precisely on the mega-caps every value book holds. Count-weighted coverage had looked healthy throughout; the lesson we record is that *coverage audits must be value-weighted*. The fix: the top 250 unmapped CUSIPs by value (65% of the hole) hand-mapped and verified individually; 61 new price histories fetched; caches stitched; non-ETF value coverage raised from 63% to 85.4%, with a documented ladder to $\sim 93\%$ (identifier-service tail) and $\sim 98\%$ (CRSP-tier, delistings).

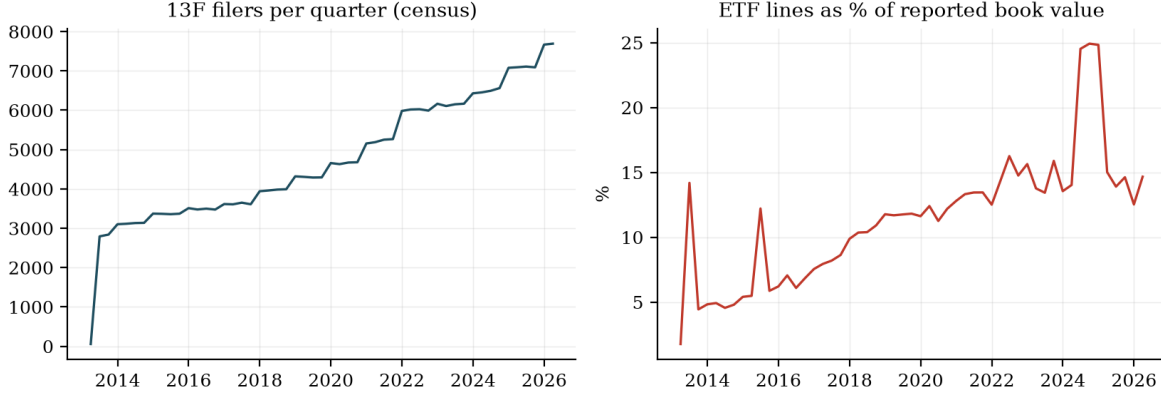


Figure 2: Left: 13F filers per quarter (census). Right: ETF lines as a share of reported book value — material and growing, and measured to be innocuous for ranking signals (uniform, not differential, contamination).

2.5 The re-verification campaign and one retraction

Because coverage selection could in principle manufacture or hide results, every material verdict was re-run on the expanded universe: 24 re-runs. The pre-registered expectation was that the hole was quasi-random with respect to signals (EDGAR name formatting does not predict returns), so levels could move where mega-caps matter but verdicts should stand. Twenty-one verdicts were unchanged; two instruments *strengthened* with completer books (the flow instrument’s selling contrast from $t = +13.8$ to $t = +20.9$; synthetic short interest’s correlation with actual short interest from 0.47 to 0.55); NCB replicated inside its historical band and survived two random 50%-coverage halvings ($t = +3.45 / +2.79$), the generic proof that quasi-random coverage cannot flip verdicts.

One result flipped and is retracted: a daily overlay claiming that a synthetically detected “stressed” manager’s top holdings underperform over the following five days ($t = -2.16$ on the old universe) went to $t = +0.37$ on complete books and is dead even inside its own sub-window. The diagnosis is an eligibility-selection artifact: coverage filters had excluded managers holding unmapped mega-caps, biasing the manager sample toward a stratum where the effect appeared. The research funnel had already declined to promote the result (net returns too thin); the audit converted “not promotable” into “false.” We keep both the machinery (reused by the risk monitors of Section 10) and the negative finding.

3 The manager universe: a census, not a chosen list

The filer universe is the full census of 13F filers each quarter — 2,793 in 2013Q2 growing to 7,693 by 2026Q1 (Figure 2) — and this is a deliberate result, not a default. Five families of eligibility filters were tested against the census with paired quarter-by-quarter deltas on identical stocks: activity and concentration screens (four variants), long-horizon “patient” managers, top-past-performer screens, passive-manager exclusions, and ETF purges. Every filter either hurt (activity screens up to $t = -2.54$; patient screens $t = -2.35$ to -3.25) or did nothing. The votes discarded with the managers carried information; the electorate is not the dial that matters (Section 6.3 shows the dial that does — the definition of the vote itself).

The ETF and passive-manager question. Because index products are 12–15% of reported book value (Figure 2, right), we closed this question on three independent fronts rather than by assumption. (i) *Tradable universe*: of 7,886 audited ETF instruments, one appears in the crosswalk and none in the price panel — ETFs were never buyable in any backtest. (ii) *Inside books*: purging ETF lines from filer books changes the conviction threshold of $\sim 25\%$ of managers per quarter and releases 17% more conviction votes (single-stock bets hidden behind index positions), yet the stock-level ranking is unchanged (rank correlation 0.974; top-100 overlap 97/100; paired delta $+0.01\%$ /quarter, $t = +0.08$): uniform contamination does not distort a ranking signal. (iii) *Passive managers as voters*: only ~ 10 filers per quarter have active share below 25% — the giant index complexes, one third of aggregate AUM — and they produce $\sim 1\%$ of conviction votes; removing them moves the spread by six basis points ($t = +1.19$). We summarize the asymmetry as *elephants without votes*: cutting active managers hurts, cutting passive ones is noise, and the census stands on measured parsimony.

Tradable asset universe. Common equities with a mapped current ticker and price $\geq \$1$; no ADV floor at the signal-evaluation layer (the illiquid short side is where several literature effects live), with liquidity entering explicitly through each signal’s denominator or the engine’s cost model rather than through silent universe truncation. The production engine (Section 8.3) applies its own stricter tradability filter ($\$4$ price, $\$5\text{M}$ median dollar volume, 252 days of history).

4 Evaluation protocol and research discipline

All signal evaluations share one protocol, fixed before any return was examined. Decisions at quarter-end $+45$ days (justified in Section 2.3); holdings via the point-in-time store; quintile portfolios, equal-weighted primary and value-weighted reported; Q5–Q1 spreads on forward quarterly returns; Newey–West t -statistics (4 lags); Spearman rank information coefficients; a $\$1$ price floor; ties broken by seeded random jitter (deterministic alphabetical tie-breaking is an artifact generator where sorts have mass points at zero). Cumulative-return figures are log-scale throughout.

Two statistical conventions are worth stating precisely. Forward quarterly returns are *sums of daily simple returns* — a log-return approximation that ignores intra-window compounding. It is applied identically on both sides of every sort, so spreads are quintile-neutral to the convention; reported levels are mildly conservative for high-volatility names. Newey–West with four lags on *non-overlapping* quarterly series is deliberately conservative: the spreads carry no mechanical overlap, so the correction can only widen standard errors against residual autocorrelation and volatility clustering.

Why equal weights as the measurement base. The evaluation question is cross-sectional: *does the ranking order next-quarter returns?* Under that question every name is one observation, so the diagnostic portfolio weights names equally — value-weighting re-asks the question “does it order returns among mega-caps?”, a narrower hypothesis whose spread variance is dominated by a handful of names (the effective N collapses, and with it the fundamental-law arithmetic $\text{IR} \approx \text{IC}\sqrt{N}$ that our signals live on). EW is therefore the *measurement* convention and cap-weighting the *implementation* convention: the tradability question EW raises (small-cap loading) is answered downstream and explicitly — the engine trades a filtered universe with costs (Section 8.3), and the long-only product is built on cap weights with a tilt (Section 8.4). VW variants are reported throughout as the bridge.

Ex-ante neutralization: the dart-throwing-manager test. Whether a signal is residualized on size and liquidity is decided by a formal placebo, not taste. We simulate value-weighted books held by uninformed (“dart-throwing”) managers and ask whether the candidate signal’s correlation with log market cap and log dollar ADV arises mechanically in that world. If it does, the headline version of the signal is the cross-sectional residual on (log mktcap, log ADV), fixed *before* looking at returns; if it does not, the raw version is the headline. Both versions are always reported. This single rule prevents the two symmetric errors — repackaging a size tilt as information, and destroying information by blanket neutralization — and the catalogue delivers one clean exhibit of each. NCB *raw* is nothing ($t = -0.9$ equal-weighted): the mechanical large-cap loading of a counting signal drags against its information in this mega-cap-led sample, and residualization *unmasks* the factor ($t = +3.5$) rather than manufacturing it — the residual is not a tweak, it is where the signal lives. Days-of-ADV is the mirror image: raw it replicates its published value-weighted point (+1.46%/month), residualized it dies ($t = -0.4$) — the published “crowding effect” *was* a size/liquidity exposure. One rule, two opposite unmaskings, both decided before returns were seen.

Verdicts are paired deltas. Whenever two designs are compared (a filter versus the census, a conditioning versus unconditional, combiner A versus B), the verdict statistic is the quarter-by-quarter difference on the common sample with its own Newey–West t — never two headline Sharpe ratios compared by eye.

Pre-registration and multiplicity. Experiment families declare hypotheses, target cells, expected signs, and thresholds in the script docstring before running. Families of five tests carry a Bonferroni bar of $|t| \geq 2.57$ (stated with the iteration count wherever invoked); results between 1.96 and the bar are labeled suggestive and never promoted. Post-hoc entrants — however well they perform — are labeled candidates: this rule is applied against our own late results, including an IPCA combiner that beats the production model’s point estimate (Section 8.2).

Smoke never decides. Short-panel runs exist to catch crashes. Four short-panel mirages inverted or vanished on the full panel during this project — including a correlation-triggered stress variant that looked excellent in a 10-quarter smoke and inverted at 50 — so no reduced-sample number is ever quoted as evidence.

Loud failure. A parse recovering zero rows against a cover page declaring entries is an error, not an empty filing; a missing quarter raises; every data-quality decision is counted in a rejection ledger that regenerates the data documentation. The same standard applied upward produced the retraction of Section 2.5.

5 A taxonomy of positioning signals

Everything tested falls into six classes. The class structure is not decorative: outcomes are homogeneous *within* class and the two laws of Section 12 explain the pattern *across* classes.

A. Conviction and positioning changes. Counts of costly actions: new top-of-book entries (NCB), top-decile holdings growth, breadth changes, entry/exit rates. *Where the alpha lives.*

- B. Flow and forced selling.** The manager-level implied-flow instrument and its stock-level projections: forced-sale supply (FSS), flow-driven demand pressure, synthetic short interest. *The mechanism engine; prices as a regime-dependent short.*
- C. Crowding and ownership states.** Levels: days-of-ADV held, ownership share, holder concentration, institutional-ownership changes, factor crowding, co-ownership graphs, latent-strategy crowding. *Fifteen formulations, zero predictors — Law 1.*
- D. Smart money and universe selection.** Performance-weighted votes, skill scores, manager filters. *Zero of nine, with the root cause measured (skill persistence 0.25).*
- E. Latent structure.** ICA on flow panels, NMF on portfolio weights. *Valid identification instruments; modest tradable edge, reported as such.*
- F. Event-clock and tactical.** Ragged per-name entry, quarter-end effects, amendment events, daily stress detection, monitors. *The clock studies; one anti-panic rule and one risk conditioner survive.*

5.1 The seventeen-signal catalogue

Class A–C constructions were evaluated jointly as a pre-registered catalogue of seventeen signals under the Section 4 protocol, each with a literature anchor where one exists, each evaluated raw and residualized with the headline version fixed ex-ante by the dart-throwing test. Figure 3 shows the information coefficients; Table 2 the headline statistics. Three facts matter. First, the ordering: the only $|t| > 2.5$ entry is NCB, and the runner-up is its sibling (top-decile conviction level). Second, the replication record: where published numbers exist, ours match — breadth change replicates Chen, Hong and Stein [2002] at +1.65%/quarter; institutional-ownership level and share held are dead after size, as Gompers and Metrick [2001] would predict once their size exposure is removed and as Chincarini et al. [2020] report; days-of-ADV replicates its published raw value-weighted point (+1.46%/month) and dies residualized — the published effect *is* a size/liquidity exposure. Third, the confidential-treatment reveal group (Agarwal et al. [2013]) does not replicate on our window (−0.21%/quarter, $t = -0.86$).

Three statistical notes discipline the table. *Multiplicity*: seventeen signals in two versions are 34 tests; the catalogue-wide Bonferroni bar at the 5% level is $t \geq 3.2$, and NCB is the only entry that clears it ($t = 3.54$) — though its promotion rests on the implementation replications and the hyperparameter surface (Section 6), not on any single t -statistic. *Magnitude*: an IC of 0.023 looks small until the fundamental law of active management prices it — $IR \approx IC\sqrt{N}$ with $N \approx 1,900$ names per quarter gives ≈ 1.0 , which is the Sharpe observed; cross-sectional breadth, not per-name accuracy, is where 13F information lives. *Weights*: value-weighted variants in the module reports use the shares-inferred capitalization panel whose corrupt tail is documented in Section 8.4; the equal-weighted headline numbers involve no weights and are unaffected, and VW columns should be read as indicative only.

5.2 Replications, stated as such

We list the replication record in one place because it calibrates trust in the new results. Chen, Hong and Stein [2002] breadth: replicated. Gompers and Metrick [2001] ownership level: the level effect does not survive size control in the modern sample, consistent with the mechanism being institutional *flows* into preferred characteristics rather than the level itself. Days-of-ADV (Chincarini

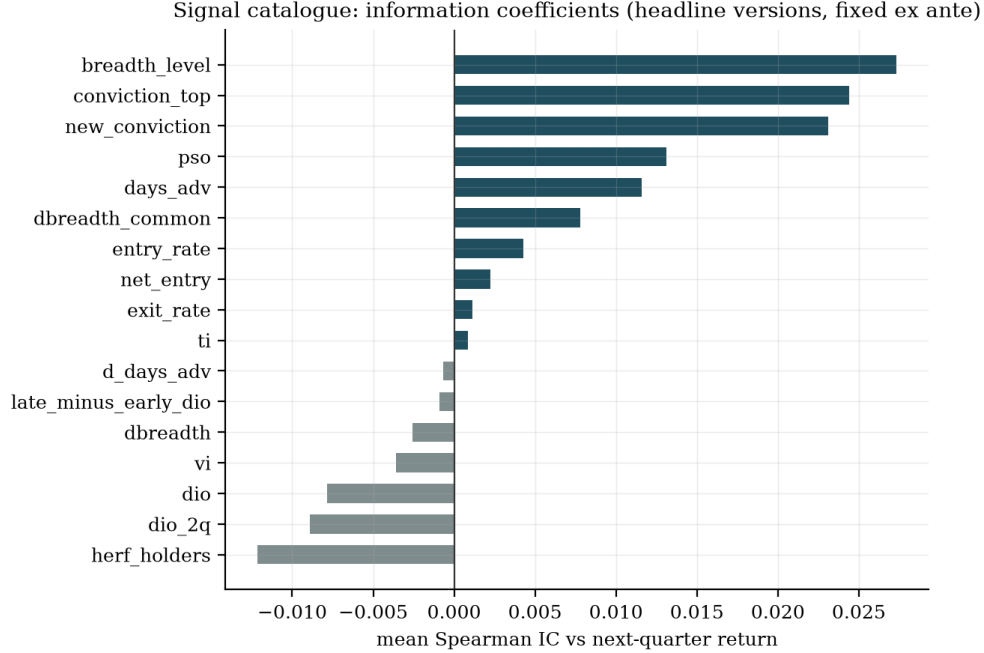


Figure 3: Mean Spearman information coefficients of the catalogue (headline versions, fixed ex ante). The conviction family leads; state variables cluster at zero or negative.

et al. [2020] and related): raw point replicated, removed by residualization. Agarwal et al. [2013] confidential holdings: not replicated on 2013–2025. Miori and Cucuringu [2022] position-imbalance reversal: evaluated in its oracle form and after bias correction (Section 10); the adjusted premium is materially smaller. Coval and Stafford [2007]: the mechanism (redemption-driven selling) replicates with $t = +20.9$; the expected-flow *forecasting* construction does not transfer to 13F-implied flows (Section 7). Kelly, Pruitt and Su [2019] IPCA and the bootstrap selection protocol of Oliveira, Guzman and Firoozye [2025] are applied as methodology, each with its own verdict.

6 New-Conviction Breadth (NCB)

6.1 Mechanism

A 13F reveals almost nothing about why a position exists — except in one corner. When a manager *opens or materially grows* a position that lands in the *top decile of their own book*, the filing records an action with three properties: it was expensive (top-decile room is made by displacing existing conviction), it was deliberate (indexing, rebalancing, and inflows do not create fresh top-of-book positions), and it is comparable across managers (each book’s own weight distribution defines its threshold). NCB counts, per stock and quarter, the fraction of filers that took this action:

$$\text{NCB}_{i,t} = \frac{1}{N_t} \# \left\{ m : w_{m,i,t} \geq q_{0.90}(w_{m,\cdot,t}), \quad s_{m,i,t} \geq 1.25 f_{i,t} s_{m,i,t-1} \right\},$$

where w are within-book weights, s share counts, and $f_{i,t}$ a per-instrument split factor estimated from the holders themselves: the modal share ratio across a stock’s continuing holders identifies splits exactly (untouched holders’ ratios all land on the split atom), validated 6/6 on prominent splits. The detector’s false-positive rate has not been separately audited; its thresholds are deliberately

Table 2: Signal catalogue, headline versions, 49 event quarters. Spread is the EW Q5–Q1 quarterly spread; t is Newey–West; “resid” marks signals whose headline is the size/liquidity residual by the ex-ante placebo rule.

signal	version	spread/q	t	Sharpe	IC	mono
NCB (new_conviction)	resid	+0.0218	+3.54	1.08	+0.023	0.58
conviction_top	resid	+0.0223	+2.39	0.65	+0.024	0.58
breadth_level	resid	+0.0193	+1.48	0.36	+0.027	0.55
late_minus_early_dio	raw	+0.0068	+1.24	0.34	−0.001	0.53
entry_rate	raw	+0.0134	+1.23	0.36	+0.004	0.57
dbreadth_common	raw	+0.0049	+1.07	0.27	+0.008	0.50
exit_rate	raw	+0.0112	+0.89	0.24	+0.001	0.54
turnover imbalance	raw	+0.0031	+0.56	0.13	+0.001	0.51
value imbalance	raw	+0.0016	+0.37	0.09	−0.004	0.53
net_entry	raw	+0.0019	+0.33	0.07	+0.002	0.51
dbreadth	raw	+0.0022	+0.32	0.06	−0.003	0.53
Δ days-ADV	resid	+0.0016	+0.28	0.06	−0.001	0.51
days-ADV	resid	−0.0066	−0.41	−0.12	+0.012	0.50
pct. shares held	raw	−0.0068	−0.74	−0.19	+0.013	0.52
Δ inst. ownership	raw	−0.0041	−0.76	−0.22	−0.008	0.45
holder Herfindahl	resid	−0.0048	−1.09	−0.29	−0.012	0.47
Δ IO (2q)	raw	−0.0083	−1.89	−0.46	−0.009	0.48

conservative (modal ratio at least 15% away from one, held by 10%+ of continuing holders or 30+ filers with $2.5\times$ dominance over the runner-up atom), so the failure mode is a missed split — which surfaces as an outlier position change — rather than a fabricated one. Split adjustment matters because unadjusted splits are *differential* contamination — they fabricate “buys” precisely in the names that moved most — unlike ETF contamination, which is uniform and measured innocuous (Section 3).

The dart-throwing placebo fires for NCB (uninformed value-weighted books mechanically open more new positions in large liquid names), so the headline is the cross-sectional residual on log market cap and log dollar ADV, re-ranked. The stock-level universe is every name with at least five holders, *including zero-vote names* — evaluating only voted names was an early bug worth recording, since the zeros carry the sort’s information too.

6.2 Results and replications

On the expanded universe, 2013Q3–2025Q4: quintile spread +1.9%/quarter, $t = +3.44$, Sharpe ≈ 1.0 , IC +0.023. Across two universes (pre- and post-coverage-fix) and four independent implementations (catalogue spec, standalone module, interaction-module replication, engine panel), the t -statistic lands in a 3.3–4.0 band eight times. These are *implementation replications* — robustness to construction and coding choices on the same data, not independent samples — and are claimed as exactly that. The signal survives two random 50%-coverage halvings ($t = +3.45/+2.79$). Attribution against market, size, momentum, reversal, and illiquidity factors leaves the premium intact (the joint Fama–MacBeth λ of Section 8.1, $t = +3.04$ holding nine other characteristics, is the strictest version of this statement), and no IPCA latent factor instrumentable by the ten characteristics absorbs it (Section 8.2).

6.3 Hyperparameters: an a-priori spec and a measured surface

The specification (book-weight quantile 0.90, growth 1.25, minimum holders 5) was fixed a priori and was the only combination ever run during the research phase — the eight replications carry no tuning bias. A bootstrap-robust selection study in the Oliveira, Guzman and Firoozye [2025] protocol then mapped the full $4 \times 3 \times 3$ surface: in-sample 2013–2020, coupled moving-block bootstrap (block 4, $B = 2,000$, common resample indices across configurations), selection on the 5th percentile of the bootstrap Sharpe, frozen and evaluated 2021–2025. Three findings (Figure 12, left; full table in Appendix C):

1. **The surface is uniformly positive out-of-sample:** all 36 configurations post OOS Sharpe between +1.05 and +2.53 (every OOS $t \geq 3.2$). The default sits mid-surface; the signal is not fragile to its parameters.
2. **The gradient is monotone toward sharper conviction** (higher weight quantile, higher growth bar), in-sample and out-of-sample — mechanism-consistent, and implying the a-priori default was conservative. The sharper corner is a post-hoc candidate under the funnel rules, not a new specification.
3. **An honest null on the selector:** the robust 5th-percentile statistic picked the same configuration as the naive in-sample maximum and ranked the OOS ordering slightly worse (rank correlation +0.41 vs +0.53); at thirty in-sample quarters the percentile tracks the point estimate. The bootstrap’s contribution here is distributional context (even the default’s 5th percentile is non-negative), not selection.

6.4 A machine-learning cross-check that found the same thing

As an ex-post exercise, we trained ridge, k -NN, random-forest and gradient-boosted regressors on the twelve change-class variables to predict the next quarter’s market- and size-residualized excess return (cross-sectional rank z -score target; per-quarter winsorize/ z -score features; strictly expanding walk-forward). Every architecture predicts the size-hedged cross-section out-of-sample ($t = +2.6$ to $+3.4$ on the residualized target) — and the exercise is a third independent route to the same conclusion: NCB alone beats the twelve-variable model (+3.12%/q, $t = +3.50$ vs +2.45%/q, $t = +3.18$; paired $t = -1.09$); the prediction is 0.52 rank-correlated with NCB; orthogonalized to NCB the model retains nothing ($t = +0.40$); and the boosting’s own permutation importance ranks NCB first by a factor of three. There is no nonlinear premium (boosted-minus-ridge paired $t = +0.27$): the linear production model of Section 8.1 already extracts what exists. The exercise also yields a practical lesson: evaluated on *raw* returns the same predictions lose money — the prediction inherits a small-cap exposure from the raw count features, and in a mega-cap regime that exposure dominates. Positioning factors must be traded size-hedged; NCB’s residualized construction does this by design.

7 Forced-Sale Supply (FSS)

7.1 The flow instrument

For each filer, implied quarterly flow separates client/leverage money from market growth of the book:

$$\text{flow}_{m,t} = \frac{\text{AUM}_{m,t}}{\text{AUM}_{m,t-1}} - (1 + r_{m,t}^{\text{book}}),$$

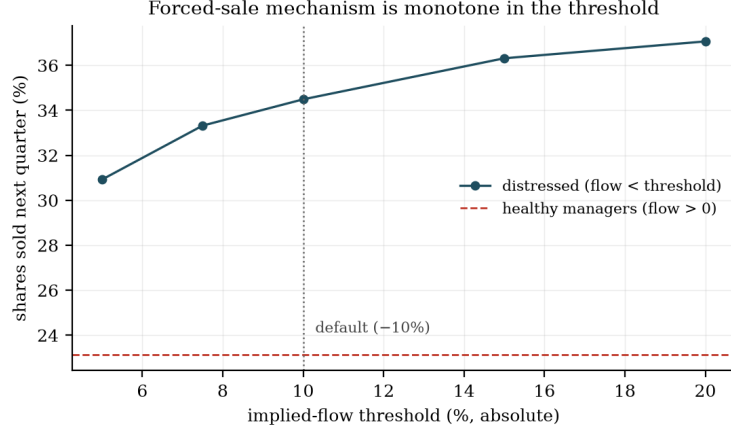


Figure 4: The forced-sale mechanism is monotone in the flow threshold; the a-priori default is not a knife edge. No returns are involved in this figure.

with r^{book} the frozen prior book marked on the price panel. The instrument is validated against subsequent *behavior*, not returns: managers with flow $< -10\%$ sell 34.5% of held shares over the next quarter versus 23.1% for managers with positive flow ($t = +10.6$; the contrast against the strictest healthy definition reaches $t = +20.9$). Completer books strengthened the instrument (the campaign of Section 2.5), as a real instrument should. Selling by distressed managers is approximately pro-rata across the liquidity spectrum (liquidity-ordering correlation -0.02): the folklore “they sell the liquid names first” is not in this data.

7.2 Threshold justification without selection

The -10% cut was set a priori. Its justification is the *shape* of the mechanism, which involves no selection at all: the sold-fraction delta versus healthy managers rises smoothly through every threshold — $+7.8\text{pp}$ at -5% , $+11.4\text{pp}$ at -10% , $+13.9\text{pp}$ at -20% , each with t between $+9.6$ and $+11.4$ (Figure 4). The default is a point on a monotone curve, chosen where purity meets sample size (144 distressed managers per quarter, versus 65 at -20%).

7.3 From managers to stocks, and what the price says

Stock-level supply aggregates each distressed manager’s prospective liquidation over average daily dollar volume: $\text{FSS}_{i,t} = \sum_{m \in \text{dis}} |\text{flow}_{m,t}| \text{AUM}_{m,t} w_{m,i,t} / \text{ADV}_{i,t}$. Pricing this basket honestly required a benchmark audit that we report as a finding in its own right. Against the universe *mean*, the basket “underperforms” by $3.8\%/ \text{quarter}$ — but so does nearly everything: the cross-sectional mean of quarterly stock returns sits $\sim 4\% / \text{quarter}$ above the median through small-cap skewness, and early internal notes quoting a $-4.5\% / \text{quarter}$ cohort effect against the mean overstated the alpha. Against the median the basket is flat. The tradable statement is the index-hedged one: approximately flat 2013–2020 (Sharpe -0.1 to -0.5 across all fifteen threshold $\times$ basket configurations) and **paying $+0.6$ to $+1.0$ Sharpe in 2021–2025** (t up to $+4.0$) at *every* point of the grid. The grid moves as one block: no configuration is special, none is fragile, and what the surface identifies is a *regime* — the degrossing/rates era — rather than a hyperparameter.

The same surface furnishes a methodological result we consider as valuable as the factor itself: the in-sample-to-out-of-sample rank correlation across configurations is *negative* (-0.81 for the

bootstrap 5th-percentile statistic). Under a regime flip, any in-sample selector — robust or naive — is anti-informative. The Oliveira, Guzman and Firoozye [2025] machinery assumes a stationary utility distribution; NCB’s surface satisfies it (rank correlation +0.53) and selection works there; FSS’s does not, and the correct use of the protocol is to refuse to select and report the regime (Figure 12, right). We apply our tools critically or not at all.

7.4 Prediction fails at every clock (Law 2)

Three designs tried to move FSS from observation to anticipation, and all failed their mechanism gates, which is why the production leg reacts to filings rather than front-running them. (i) *Quarterly flow forecasting* in the Coval and Stafford [2007] expected-flow form (lagged flow + performance + convexity, expanding coefficients): 13F-implied flow has no one-quarter-ahead predictability (OOS rank correlation -0.015 , $t = -1.21$); predicted-distressed managers do not sell more than predicted-inflow managers ($t = +0.69$, against $t = +20.9$ contemporaneous). The literature forecasts *reported* mutual-fund TNA flows; 13F-implied flow adds leverage, unreported assets, and frozen-book error, which destroys the autocorrelation. (ii) *Daily synthetic stress*: retracted (Section 2.5). (iii) *Single-name crash cascades*: the middle link (salient position crash $\rightarrow$ next-quarter redemption) does not clear its bar ($t = -1.47$); redemptions respond to aggregate performance, which the flow instrument already captures contemporaneously.

8 Combining models and net-of-cost production results

8.1 The Fama–MacBeth production combiner

The production question — one ranking merging the surviving 13F signals with classical characteristics — was raced across three combining schools on identical quarters, strictly out-of-sample (expanding estimates, twelve-quarter warm-up). Predictors, pre-registered: size, momentum, beta, low-volatility, short-term reversal, liquidity, plus NCB, breadth change, FSS, and flow pressure, all as centered cross-sectional percentile ranks.

- **A. Fama–MacBeth/Lewellen** (per-quarter cross-sectional OLS premia, expanding mean, $E[r] = \bar{\lambda}'x$): spread +5.4%/q ($t = +2.65$), **Sharpe 0.86**, IC +0.030.
- **B. IC-weighted composite** (Grinold–Kahn): +4.3%/q, Sharpe 0.69 — univariate weights ignore signal overlap.
- **C. Naïve rank average** (NCB + FSS only): +2.6%/q, Sharpe 0.74. Paired deltas: A–B $t = +1.18$; A–C $t = +1.96$.

The joint-marginality table is the deliverable the one-at-a-time factor ladder cannot produce: holding all ten predictors simultaneously, NCB’s premium is $t = +3.04$ and breadth change’s $t = +4.01$ — the 13F premia are marginal to six classical characteristics at once. Breadth change is the canonical example of an *ingredient*: weak as a standalone quintile sort (Table 2) yet the second-strongest joint premium; signals correlated with controls reveal their value only in the joint estimate. Figure 5 shows the race cumulatively.

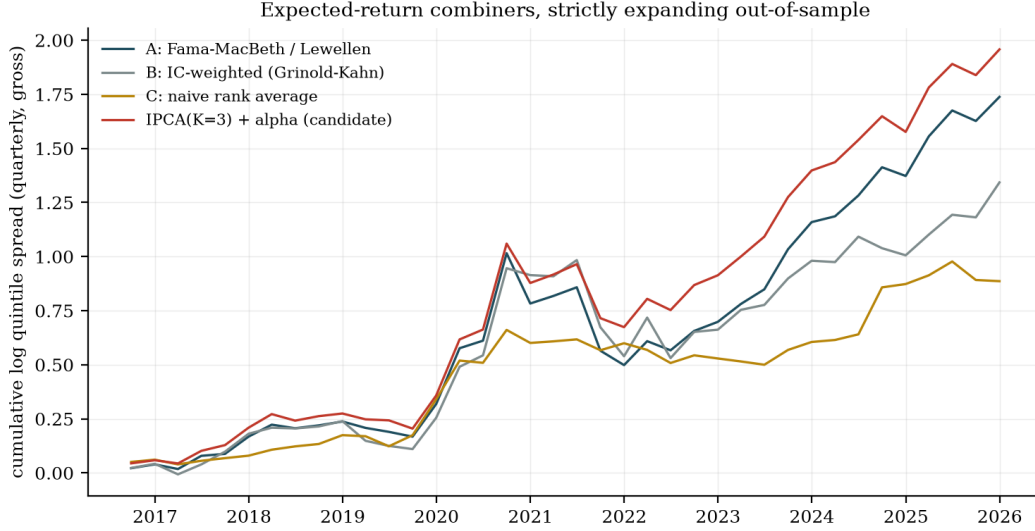


Figure 5: Expected-return combiners, strictly expanding OOS, gross quintile spreads (log cumulative). The Fama–MacBeth combiner is the production model; the IPCA challenger is a candidate under the funnel’s promotion rules.

8.2 The IPCA challenger: is the premium a latent-factor loading?

Under Kelly, Pruitt and Su [2019], factor loadings are instrumented by characteristics ($\beta_{i,t} = z'_{i,t}\Gamma$) and estimated jointly with latent factors by alternating least squares; our implementation was verified line-by-line against the authors’ reference package (N_t -weighted Γ step, intercept as pre-specified factor, W_α wild bootstrap with $t(5)$ multipliers, 500 draws). Fifty quarters, instruments = constant + the ten predictors. Results: total R^2 0.10–0.12 at $K = 1..4$; the formal W_α test does not reject $\Gamma_\alpha = 0$ ($p = 0.41$ – 0.59 — expected at $T = 50$; the test needs hundreds of periods for power); the interpretable spanning statistic — the time-series alpha of the NCB-managed portfolio on the latent factors — is $t \approx +6$ at every K . Out-of-sample, the unrestricted forecast $z'(\Gamma_\beta\bar{\lambda} + \Gamma_\alpha)$ posts Sharpe 0.97 and beats the FM combiner paired ($+0.55\%/q$, $t = +2.08$); the genuinely rank-restricted forecast ($z'\Gamma\bar{\lambda}$) ties FM ($t = -0.75$).

Two disciplines govern the reading. Algebraically, the unrestricted unconditional forecast is *not* rank-compressed (Γ_α is a free vector, so $\Gamma_\beta\bar{\lambda} + \Gamma_\alpha$ spans $\mathbb{R}^{11}$): the gain over FM comes from a different, pooled, structure-regularized estimator of the same premia — not from “compression keeping the alpha,” and the leg that is genuinely compressed confirms this by not beating FM. Procedurally, IPCA entered after the race was scored: a post-hoc fourth entrant at $t = +2.08$ does not clear the pre-registered promotion bar (2.57), so **FM remains the production combiner** and IPCA+alpha is recorded as the best point estimate on the protocol, pending fresh sample. What survives unconditionally is the answer to “is NCB beta in disguise?": no latent factor instrumentable by these characteristics absorbs it.

8.3 The net engine

Research spreads are gross and frictionless by construction; the engine is neither. Every panel — survivors and rejected signals entered *inverted*, so a positive net Sharpe means “the research conclusion pays” — runs through one realistic long-short book: tradability filters (\$4 price, \$5M median dollar volume, 252 days of history), 20% tails with a 40-name minimum per leg, rebalanced

Table 3: Production-engine results, net of 5 bps/side commissions and 50 bps p.a. borrow on short legs; expanded universe; tradability filters \$4 / \$5M ADV / 252d. “cost” is the measured annualized commission drag (net run minus zero-fee run); “turn” the implied one-way annual turnover (traded notional over book, per year). Research-rejected signals enter inverted, so a positive Sharpe means the research conclusion prices.

strategy	ann. net	vol	Sharpe	maxDD	Calmar	hit	cost (bps/y)	turn
NCB (residualized)	+2.9%	6.7%	+0.42	-18.7%	+0.15	51%	65	6.5x
FM score (10 predictors)	+4.0%	16.9%	+0.24	-62.1%	+0.06	51%	43	4.3x
NCB+FSS book	+6.2%	6.0%	+1.02	-10.6%	+0.58	53%	67	6.7x
FSS short leg	+10.4%	12.2%	+0.85	-26.1%	+0.40	54%	65	6.5x
dBreadth (common filers)	+1.4%	11.1%	+0.13	-28.2%	+0.05	53%	61	6.1x
NCB v2 (catalogue spec)	+2.6%	8.3%	+0.31	-26.2%	+0.10	53%	50	5.0x
Conviction top (level)	+2.5%	12.3%	+0.20	-49.6%	+0.05	52%	33	3.3x
ICA reversal	-2.1%	8.2%	-0.26	-24.7%	-0.09	49%	51	5.1x
SSI (inverted)	-2.0%	9.9%	-0.20	-34.8%	-0.06	48%	29	2.9x
NMF crowding (inverted)	+0.7%	11.9%	+0.06	-51.4%	+0.01	53%	45	4.5x

at each signal’s own decision dates, **5 bps per side commissions and 50 bps p.a. borrow on short legs**, daily net returns.

The cost assumptions are justified by the trading profile rather than asserted. Five basis points per side is in line with realized all-in commission-plus-spread costs for institutional trading of liquid US equities, and this strategy is the easy case: quarterly rebalancing (measured one-way turnover of 3–7× per year, Table 3), no intraday urgency, and an explicit liquidity control — the tradability filter plus the per-leg breadth minimum keeps individual positions small relative to ADV, which is what keeps slippage beyond the flat fee second-order. Market impact matters at scale and belongs to the capacity discussion, not the backtest. Borrow at 50 bps p.a. is general-collateral; because the FSS leg skews illiquid we report the hard-to-borrow sensitivity: at 200 bps the FSS leg loses $\approx 1.5\%/yr$ (Sharpe $0.85 \rightarrow 0.73$) and the combo $\approx 0.75\%/yr$ (Sharpe ≈ 0.9) — the conclusions survive. The NCB+FSS book weights the two legs’ ranks *equally by design*: the no-optimization default, chosen so the production book inherits no tuned parameter; any data-driven leg weighting would require its own surface under the Section 6.3 protocol.

Table 3 reports the full battery — annualized net return, volatility, Sharpe, information ratio against the hedged benchmark, maximum drawdown, Calmar, hit ratio — for the headline strategies on the expanded universe, and Figure 6 the cumulative net curves. The engine compresses everything (costs, the stricter universe, no shorting of the illiquid tail), the ordering matches the research layer, and the research-rejected signals inverted earn approximately zero net — the engine confirming, from the cost side, that the funnel’s rejections were correct.

8.4 Long-only implementation

A naive long-only expression — holding the long basket equal-weighted against the S&P — is condemned by the composition arithmetic of Section 11 before any signal question arises: the equal-weight tilt pays the Bessembinder tax that sank the median 13F book itself. The institutional design tilts the *benchmark*: true cap weights over the top 500 names by repaired market capitalization (10% cap), multiplied by $\exp\{2k(\text{rank}_i - 0.5)\}$ on the NCB+FSS combo rank and renormalized, quarterly, net of 5 bps/side on turnover. The multiplicative-exponential form is the standard tilt of index-provider factor-tilt methodologies (MSCI-style tilt indexes multiply parent-index weights by a positive score function): it preserves positivity and long-only feasibility by construction, leaves unscored names at benchmark weight (rank 0.5), and makes the tilt strength a single interpretable

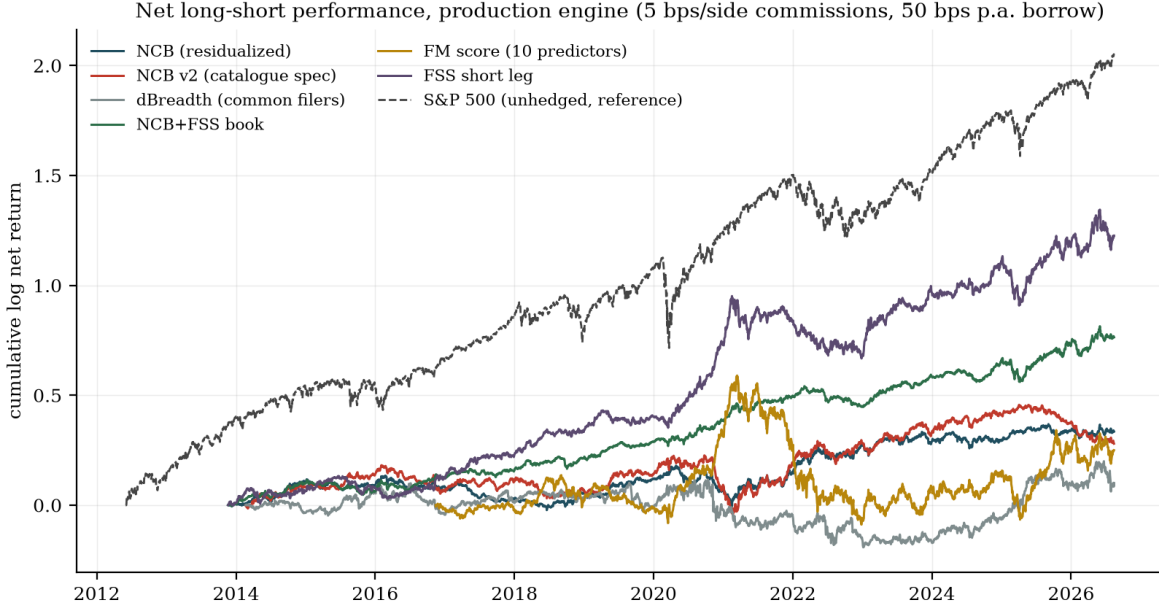


Figure 6: Net long-short performance in the production engine (log cumulative returns; 5 bps/side commissions, 50 bps p.a. borrow).

dial. The benchmark is a genuine cap-weighted model: the top 500 names by repaired market capitalization (official S&P constituent weights are licensed data; historical constituent *lists* are free, weights are not), and the repair matters — the shares-inferred cap panel is reliable as a ranked regression control but has a corrupt tail as portfolio weights, fixed by replacing $|z| > 2.5$ outliers of a cross-sectional $\log(\text{mcap}) \sim \log(\text{ADV})$ regression with their fitted values. The resulting proxy tracks the actual S&P 500 at a 2.7% tracking error with $-0.6\%/yr$ drift — close enough to serve as the cap-weighted base. Table 4 and Figure 7: the tilt adds $+1.0/+2.2/+3.6\%$ per year over the benchmark at tracking errors of 1.8/4.0/6.2% for $k = 1/2/3$ — an information ratio of 0.53–0.58, *roughly flat in the tilt strength*, the signature of a scalable signal rather than a lucky corner. All three variants also clear the S&P itself.

Justifying the tilt strength: the third surface. The one free parameter of the LO product, k , received the same bootstrap treatment as the NCB and FSS specifications (grid $k \in \{0.5, \dots, 4\}$; utility = IR of quarterly excess vs the benchmark; coupled block bootstrap of the in-sample years, one IR distribution per k ; frozen OOS 2021–2025). Figure 8: in-sample the distributions shift up with k (median IR $0.59 \rightarrow 1.09$; even the 5th percentile turns positive from $k = 1$) — and in the held-out years the ordering reverses (IR 0.42 at $k = 0.5$ down to -0.04 at $k = 4$). Two honesty caveats size this correctly. First, with 23 held-out quarters the standard error of an annualized IR is ≈ 0.4 , so the reversal is *suggestive, not proven*. Second, the mechanism is mundane: the exponential tilt is nonlinear — large k concentrates the book far from cap weights, degrading the transfer coefficient per unit of tracking error, and the 2021+ mega-cap regime taxed exactly that deviation. The natural objection — “why not re-select k each period with an expanding window?” — was simulated directly: an adaptive selector maximizing expanding-window IR picks $k = 4$ in essentially every quarter from 2015 through 2022 (in-sample always rewards aggression), which makes it approximately equivalent to holding the most aggressive tilt throughout; it earns -0.11 after 2021 against $+0.31$ for the frozen default. The defensible statement is deliberately modest:

Table 4: Long-only cap-weighted model and its combo tilts, net of 5 bps/side commissions, quarterly rebalance, 2013–2026. “Total ret.” is the portfolio’s full net return (long-only, $\beta \approx 1$: it contains the market); the *alpha* columns are “excess”/IR — against the tilt’s own benchmark in the top block, against the actual S&P 500 in the bottom block. Rel. maxDD: worst peak-to-trough of the log wealth ratio versus the same reference. No bootstrap-selected k appears *by design*: the tilt-strength surface (Figure 8) gives no evidence that selecting or re-selecting k from history helps, and weak evidence that it hurts — k is shipped as the conservative default ($k = 1$) and scaled by the investor’s tracking-error budget.

portfolio	total ret. (net)	vol	Sharpe	maxDD	excess	TE	IR	rel. maxDD
S&P 500 (actual index)	+14.8%	17.1%	0.86	−41.1%	—	—	—	—
cap-weighted 500 (S&P proxy)	+14.1%	18.0%	0.78	−44.6%	—	—	—	—
combo tilt $k = 0.5$	+14.6%	18.2%	0.80	−44.6%	+0.4%	0.9%	0.49	−2.5%
combo tilt $k = 1$ (shipped default)	+15.1%	18.4%	0.82	−44.5%	+1.0%	1.8%	0.53	−5.7%
combo tilt $k = 2$	+16.4%	19.2%	0.85	−44.1%	+2.2%	4.0%	0.57	−13.7%
combo tilt $k = 3$	+17.7%	20.4%	0.87	−43.9%	+3.6%	6.2%	0.58	−21.2%
benchmark vs S&P					−0.6%	2.7%	—	−17.0%
$k = 1$ vs S&P					+0.3%	3.6%	0.09	−13.6%
$k = 3$ vs S&P					+2.9%	7.3%	0.40	−25.5%

there is no evidence that re-selecting k helps, weak evidence that it hurts, and since large k buys tracking error without reliable additional excess, the shipped default is the conservative $k = 1$ — with k treated as the investor’s tracking-error budget (0.9% at $k = 0.5$, 1.8% at $k = 1$, 6% at $k = 3$), not as a fitted parameter. Across the three applications of the Oliveira, Guzman and Firoozye [2025] protocol, one surface supported selection (NCB: IS ranks OOS at +0.53) and two refused it (FSS −0.81, LO tilt −1.0) — the protocol’s value in this project was less in picking parameters than in *detecting when picking is illegitimate*.

The research-to-production decomposition of the FM combiner. The FM combiner’s research Sharpe (0.86, gross quintile spreads on the wide ≥ 5 -holder universe) does not survive the engine (net Sharpe 0.24, −61% drawdown), and the decomposition is a result in itself, produced by three controlled cuts. Costs are irrelevant (0.24 with commissions, 0.26 without). The engine is faithful (a plain quintile spread of the same ranking on the engine’s tradable universe reproduces the engine’s Sharpe). What kills it is composition: on the tradable universe the raw ranking retains +2.2%/q (Sharpe 0.41, $t = +1.32$), and residualizing the ranking on size and liquidity leaves *nothing* (−1.1%/q, $t = -0.70$). The combiner’s headline premium rides its two largest lambdas — size ($t = +4.45$) and liquidity ($t = +4.39$) — which pay either outside the tradable set or as risk compensation the engine cannot monetize. The tradable alpha inside the ranking is precisely its 13F components, which is why the production book is NCB+FSS (Table 3: net Sharpe **1.03**, maximum drawdown −11%) rather than the full ten-characteristic ranking. The joint FM machinery keeps its role as the *measurement* device (Section 8.1); the tradable book is built from the components that survive on the tradable universe.

9 Latent structure: ICA on flows, NMF on portfolios

9.1 Why ICA and not PCA, from the data up

The starting object is a panel: rows are quarters, columns are managers, cells are the validated implied flow of that manager in that quarter. Individual flows are not independent — common

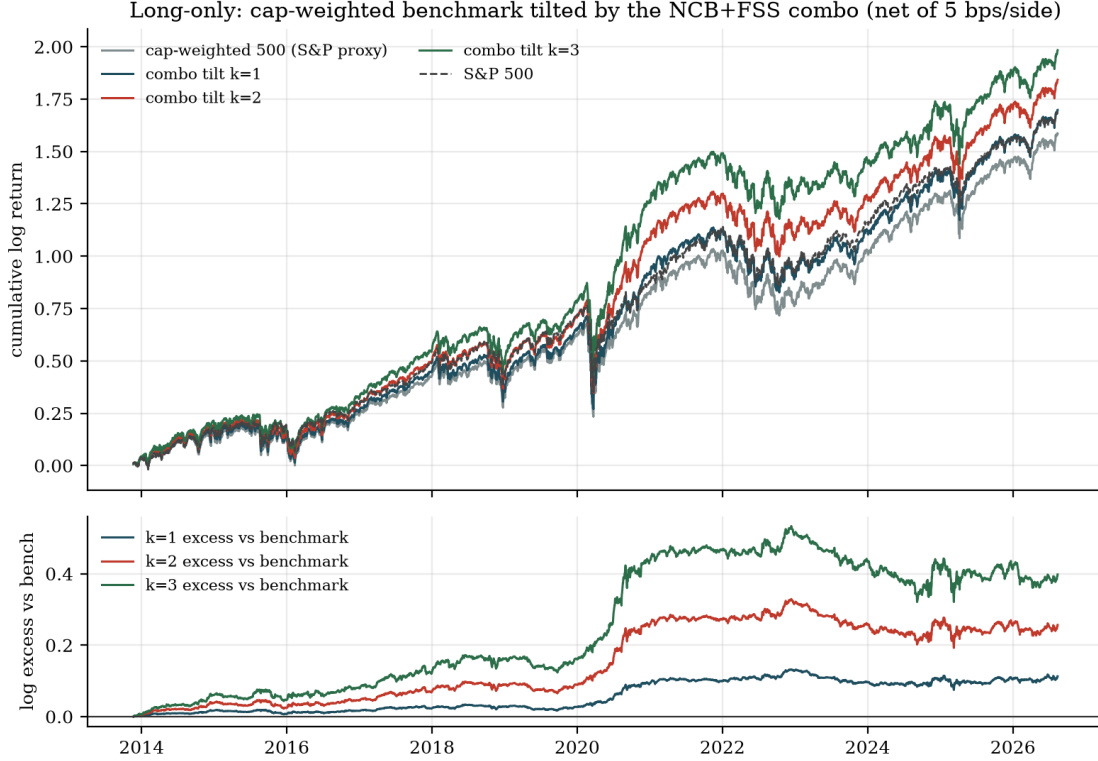


Figure 7: Long-only benchmark tilt (log cumulative, net). Top: levels; bottom: log excess versus the benchmark by tilt strength.

currents run behind them (hedge-fund redemption waves, rotation to passive, the 2020 panic) — and we want the currents separated. The situation is the cocktail-party problem: each manager is a microphone recording a mixture of conversations, and nobody hands you the conversations.

PCA cannot solve it, for a reason worth being precise about. PCA uses only covariance, and covariance is blind to *shape*: a steady $\pm$ breathing pattern and a silent-then-burst pattern with the same variance are identical to it. Worse, for Gaussian-looking data any rotation of the true sources reproduces the same covariance, so PCA returns an arbitrary rotation — a way of summing the conversations, not the conversations. The current we care about — forced selling — is precisely the bursty one: quiet for years, then one violent quarter. Its signature lives in kurtosis and asymmetry, not variance. ICA inverts the central limit theorem: mixing makes things more Gaussian, so unmixing means finding the *least* Gaussian combinations. A forced-liquidation burst is the least Gaussian object in a flow panel.

We extract $K = 6$ components with expanding-window FastICA and classify each by shape alone — high kurtosis with low persistence = forced; smooth and persistent = allocative — never touching returns in the classification. Three validations discipline the construction. *Rotation placebo*: applying the same trading logic to variance-preserving rotations of the components (what PCA would deliver) destroys the effect — reversal thesis $t = +1.29$ on ICA components versus -0.77 rotated; the information lives exactly in the shape only ICA preserves. *Unlabeled crises*: the forced components' burst quarters land on the taper tantrum, Brexit, and COVID without the model ever seeing a calendar. *Stability*: the identified subspace persists at 81% quarter over quarter (measured as subspace overlap; components may permute or flip sign without anything real changing). The tradable edge is modest and reported as modest — approximately zero net in the engine — and the

LO tilt strength: bootstrap IR distribution per k (IS 2013–2020, B=2,000, coupled) and frozen OOS

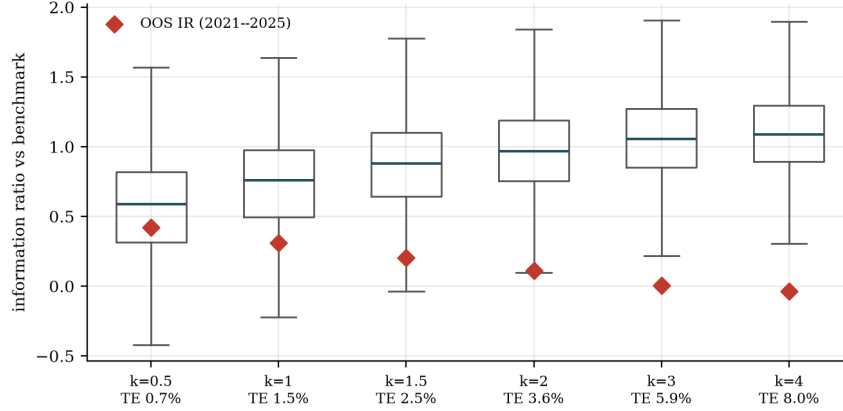


Figure 8: One bootstrap IR distribution per tilt strength (boxes: in-sample 2013–2020, coupled resamples) against the frozen held-out IR (diamonds). The in-sample ordering reverses in the held-out years — suggestive at this sample size (IR s.e. ≈ 0.4), and enough to treat the tilt strength as a risk budget rather than a fitted parameter.

piece’s value is identification: it is the cleanest demonstration in the project that a “sophisticated” method can be held to placebo and stability standards like any signal.

9.2 NMF latent strategies

Non-negative matrix factorization of the manager $\times$ stock weight panel yields components that are themselves long-only portfolios — the honest reason to prefer NMF over PCA/SVD here, since a manager is interpretable as a mixture of strategy books but not as a sum of axes with negative holdings. $K = 10$ – 12 selected by held-out manager reconstruction; the structure is stable across the coverage expansion. As a crowding predictor the construction fails (Section 12; the conditional fragility test returns the sign *opposite* to the Greenwood and Thesmar [2011] thesis), and its retained roles are measurement and clustering: the strategy mixtures are the natural pooling groups for hierarchical (James–Stein) estimation of per-manager parameters, which is where any future flow-forecasting attempt with reported-TNA data would start.

10 Event-clock studies, monitors, and tactical results

The clock itself. Entering each name on the day its information became public (per holder-filing acceptance) instead of the D+45 batch adds $+0.21\%/yr$ on the long side ($t = 0.18$) and $+0.31\%/yr$ on the short ($t = -0.22$): the cost of the batch clock is measured at approximately zero, consistent with the delay distribution of Section 2.3.

Classical interactions: five pre-registered conditional effects, none confirmed. Theory motivates conditioning each 13F signal on a classical characteristic; five such interactions were pre-registered with target cells, expected signs, and a Bonferroni bar of $|t| \geq 2.57$ for the family. None confirmed (Table 5); NCB replicated unconditionally at $t = +3.78$ inside the same module — the factor is broad, not a corner-case effect.

Table 5: Pre-registered 13F×classical interactions, 50 quarters. Verdict statistic: paired delta between the target-tercile spread and the unconditional spread, NW(4).

interaction (theory)	target	uncond. (t)	delta (t)	verdict
NCB × mktcap (limits to arbitrage)	small	+3.78	+1.34	not confirmed
dbreadth × IO (?-style)	low IO	+1.17	−1.42	not confirmed
FSS × ADV (impact where illiquid)	illiquid	−0.90	+1.05	not confirmed
pressure × vol (flow moves price in high vol)	high vol	+1.09	+0.77	not confirmed
exit_rate × momentum (exodus in losers)	losers	+0.96	+2.02	wrong sign

The pressure autopsy: why flow pressure should pay, why it mostly does not, and where the lesson survived. The economic case for a flow-pressure factor is genuine, and its failure is as informative as NCB’s success. A 13F reveals flows that were *already executed*: by the decision date the quarter’s selling has largely happened and its price impact is realized inside the filing quarter — which is why the unconditional pressure signal dies once orthogonalized to momentum, reversal, and the concurrent quarter’s return; there is nothing left to forecast. The one situation where pressure should still pay *forward* is an *unfinished liquidation*: a position large relative to ADV cannot be unwound in one quarter, so the filing reveals pending supply rather than consumed supply. This is measured, not conjectured: the mechanism stage of Section 7 shows distressed managers selling 34.5% of remaining shares in the quarter *after* the filing — liquidation demonstrably continues past the disclosure.

Two conditioning routes follow, with different fates. Route one conditions on the *stock*: pressure should matter more where volatility is high (valuation uncertainty amplifies the impact of uninformed flow) — the E4 interaction above, and the project’s cleanest example of a flicker dying properly: promising in smoke, $t = +2.06$ on the old universe (just under the promotion bar), an identification autopsy — conditioning on residual volatility versus ADV versus market cap to separate valuation uncertainty from disguised illiquidity — that returned “exploratory, no formal claim” ($t = -0.50 / -0.61$), and a fall to $t = +0.77$ on complete books: non-promotion vindicated by data it had never seen. Route two conditions on the *seller*: restrict to managers whose selling is *forced*, where continuation is guaranteed by the redemption itself rather than inferred from the stock’s characteristics. That route is FSS, and it is the one that prices. The design lesson: **condition on the agent’s constraint, not on the asset’s state** — the same asymmetry as Law 1.

Imbalance reversal (Miori and Cucuringu 2022). Replicated in oracle form and re-evaluated with the biases removed (point-in-time snapshots instead of perfect foresight of the cross-section; universe-churn control): the adjusted premium is a fraction of the published point. We report both numbers as the clearest example in the project of how much 13F results depend on who-knew-what-when accounting.

Quarter-end effects. The window-dressing design (learned manager propensity, within-loser test to kill the momentum confound, permutation placebo) rejected both pre-registered signs, with the placebo tracking the thesis — whatever exists lives in ownership/ADV structure, not in window-dressing propensity. A residual, honestly labeled hypothesis-generating: the post-photo reversal is entirely a December phenomenon (−1.91% in Q4 versus −0.05% otherwise, eleven annual observations), observationally equivalent to portfolio pumping [Carhart et al., 2002] or a January low-institutional-ownership bounce; with $n = 11$ it remains a footnote.

Amendment events. Across 250k direction-signed events, material restatements move prices *against* the direction of the correction (-0.10% at $+3d$, $t = -3.52$; clean pre-event placebo); administrative amendments are zero, as they must be; confidential NEW-HOLDINGS reveals drift positively at $+10d$ but with a contaminated placebo. The pre-registered thesis (price follows the correction) is dead; the inverted fact is consistent with amendments following stale-price attention rather than creating news.

Daily monitors. Three daily rules were evaluated against pre-registered bars on the drifted-book machinery. The short-squeeze alarm inverted into the useful rule: shorts spiking $+7\%/5d$ on doubled volume *revert* (-1.21% over the next 5d versus -0.12% calm, $t = -2.60$) — the anti-panic rule is to hold, not cover. The aggregate deleveraging index is a risk conditioner, not a signal: correlation $+0.14$ with future ten-day market volatility, zero with future returns (Law 1 again); throttling gross above its expanding 80th percentile improves the combo proxy’s Sharpe $0.49 \rightarrow 0.55$ and drawdown $-17\% \rightarrow -15\%$. Early covering at apparent exhaustion loses $1.5\%/quarter$ to save 7 bps of borrow — the pressure continues; the short is held through the quarter. The post-exhaustion tactical *long* died at its gate (stress doubles the forced-sale rate, 12.2% vs 6.7% , but the thesis-minus-placebo event return is $t = +0.95$, and the only significant cell is *continuation*): distressed books do not V-bounce.

11 Context: why imitating these books is a losing design

A descriptive base rate disciplines every 13F strategy discussion. Across 1,597 large filers ($\geq \$1B$ disclosed book, coverage-eligible, 2013–2026), the median frozen-book excess return against the S&P 500 is **-3.3% per year before fees**; only 13% of funds have a positive information ratio, and the best IR in the universe is ≈ 1.0 . Skill persistence — the rank correlation of information ratios between the sample’s halves — is $+0.25$: real but far below what manager selection requires, and the measured root cause of the entire smart-money graveyard (Section 12).

The composition arithmetic matters more than the skill statement. The cap-weighted index self-concentrates into its winners, and over this sample a handful of mega-caps generated most of the index return while the median stock lagged far behind [Bessembinder, 2018]. Any portfolio more diversified than cap-weight — and every 13F book is, by mandate, liquidity, or prudence — carries an equal-weight tilt that this regime taxed structurally. The group sorts photograph the tax directly: 27-name books lose $3.2\%/yr$ with 24% positive-IR, while 790-name books sit at 1%; true active share is monotone in IR; turnover sorts favor the active. Two design conclusions follow. A long-only-versus-S&P imitation strategy is condemned by arithmetic before any signal question arises. And the alpha in this dataset lives not in the books but in the *flows and constraints between managers* — fresh conviction and forced selling — exactly where the funnel found it. Caveats stated: the disclosed book is long-only (no shorts, cash, or intra-quarter turnover), and survivorship biases the level upward.

12 The graveyard, and the two laws it obeys

Negative results are reported with the same protocol as positive ones, because the map of what fails is half the deliverable. Table 6 lists every rejected design with its verdict statistic; each has a full module report with its pre-registration in the repository. The pattern is not random.

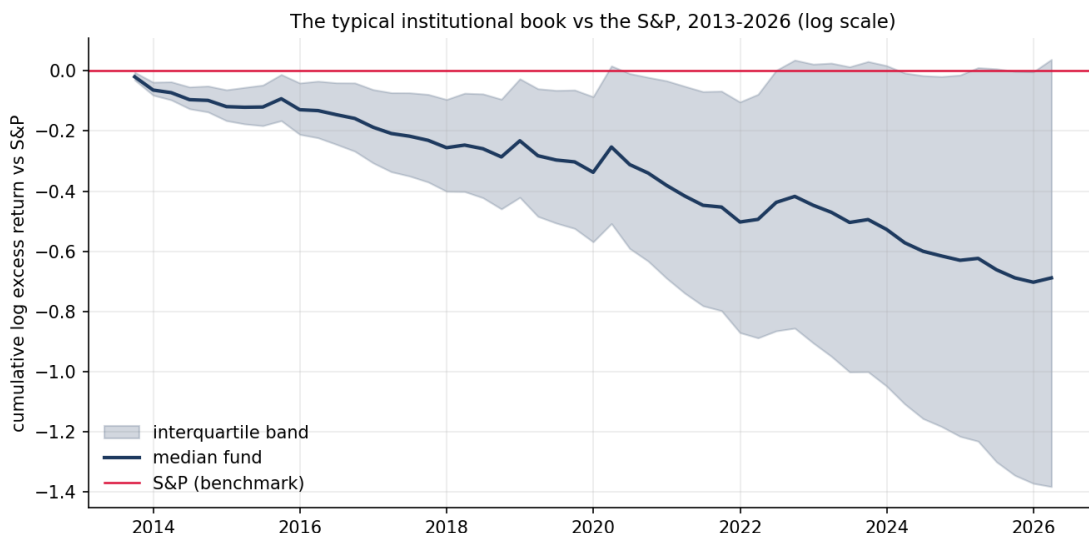


Figure 9: Cumulative log excess return versus the S&P 500 of large 13F filers (median and quantile bands): the median large institutional book loses steadily against the index before fees.

Law 1: states have no arrow. Fifteen crowding and ownership-state formulations — days-of-ADV held (level and its published value-weighted point, which replicates raw and dies residualized), ownership share, holder concentration, institutional ownership changes at one and two quarters, breadth level, factor crowding from manager tilts (three hypotheses), co-ownership graph diffusion, flow-vulnerability centrality, forced-sale absorption and overhang, latent-strategy crowding, and the aggregate stress index against future returns — carry no directional information. States earned exactly two jobs in the production stack: denominators (ADV, AUM) and risk conditioning (the deleveraging index against future volatility). The one crowding-adjacent object that pays, FSS, is not a state: it is a forced action divided by liquidity.

Law 2: distress is information when observed, never when predicted. Contemporaneous: $t \approx +10$ selling contrast, the FSS regime short, the anti-panic monitor. Predictive: quarterly flow forecasting dead at its mechanism gate; daily stress anticipation retracted; crash cascades gated; exhaustion longs dead. Both clocks, both directions, one conclusion — and it implies the production design: react fast to filings, do not front-run them.

The smart-money corollary. Nine designs weighting managers by past performance or filtering the universe by manager quality returned zero, and the base rates of Section 11 say why: skill persistence of 0.25 is too weak to select on, and manager performance rank is approximately orthogonal to *which stocks* managers hold, so reweighting the electorate reweights at random.

13 Limitations and next steps

Price panel survivorship, measured. The price vendor serves current symbols only, so the panel is conditioned on survival to the download date: delisted names are absent for their *entire* history (nothing “dies” inside a backtest window — we verified that the return construction never drops a live name mid-window). We measured the incidence and its signal alignment rather than

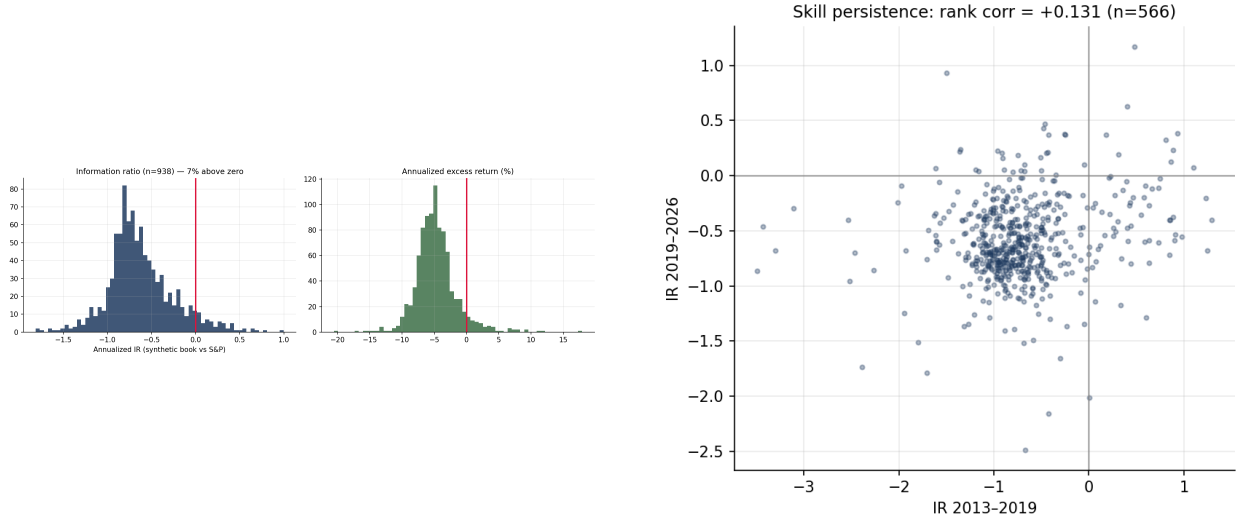


Figure 10: Left: information-ratio distribution across 1,597 large filers — 13% positive, best ≈ 1.0 . Right: skill persistence (IR rank, first vs second half): $+0.25$, the measured root of the smart-money graveyard.

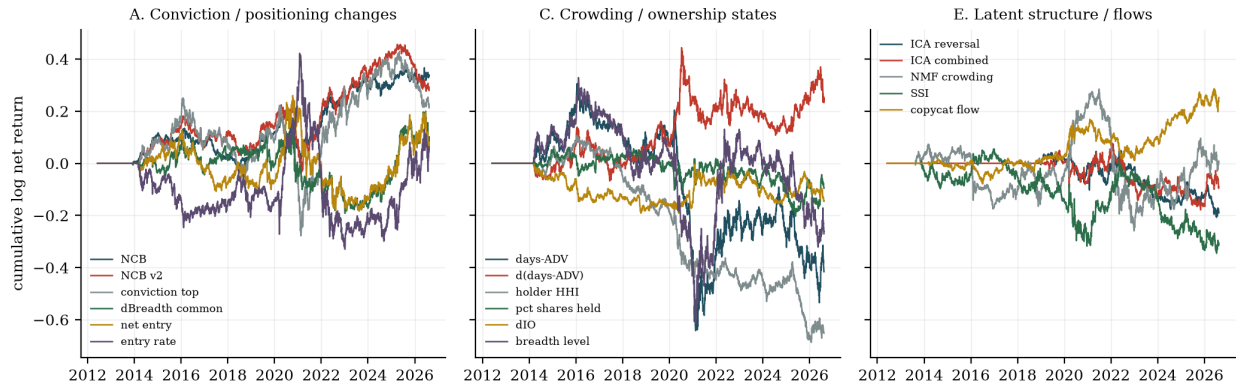


Figure 11: Net engine curves by signal class (log cumulative). The conviction class carries the performance; the state class oscillates around zero for a decade; the latent-structure class validates as measurement but earns nothing net.

asserting a direction: among universe-eligible instruments (≥ 5 holders), unmapped instruments — intentionally excluded ETFs, share classes, delisted and unresolved names — are $\sim 80\%$ of line items but $\sim 33\%$ of value, and their NCB vote rate differs from mapped names by $+0.002$ (0.0223 vs 0.0201 , $t = +2.38$): a $\sim 10\%$ relative tilt in vote intensity, small but not zero, and the honest bound on how signal-selected the tradable universe is. The economic direction of the pure-delisting component cuts *against* the measured spread: bankruptcies concentrate in the no-vote quintile (their absence flatters Q1) while acquisition targets concentrate among conviction names and exit at premiums (their absence penalizes Q5). CRSP-tier data with delisting-inclusive returns resolves both truncations and lifts value coverage from 85.4% to $\sim 98\%$ — the single highest-value data upgrade.

Coverage residuals. The identifier-service tail ($\sim 2\text{--}3\text{k}$ small CUSIPs) and a handful of throttled tickers are pending ($\rightarrow \sim 93\%$ free); delisted names are structural until CRSP.

Table 6: The graveyard. Every rejected design, its verdict statistic, and its class. Full pre-registrations and reports in the repository modules.

design	class	verdict statistic	lesson
performance-sigmoid vote weights	D	$t = +0.06$	skill rank $\perp$ holdings
top-performer universes	D	negative	same
empirical-Bayes skill weights	D	negative	same
vol-scaled performance weights	D	negative	same
activity/concentration filters (U1–U4)	D	up to $t = -2.54$	census beats filters
patient-manager universe	D	$t = -2.35 / -3.25$	same
follow-the-leader trading	B	hangover $t = -2.5$	echo arrives late
five 13F $\times$ classical interactions	A	none pass 2.57 (5 tests)	signal is broad
pressure $\times$ volatility (E4)	B	$t = +2.06 \rightarrow +0.77$	flicker died on fuller data
window dressing	F	both signs wrong	structure, not propensity
restatement direction-following	F	inverted ($t = -3.5$)	attention, not news
crash $\rightarrow$ redemption cascade	B	gate $t = -1.47$	13F clock too coarse
funding-shock beta	C	$t = +0.77$, 2–3 episodes	not yet testable
exhaustion tactical long	F	thesis–placebo $t = +0.95$	pressure continues
daily stress-anticipation short	F	retracted	eligibility artifact
correlation stress triggers	F	smoke mirage	Law: smoke never decides
cover-at-exhaustion	F	$-1.5\%/q$ for 7 bps	hold the short
hot-ownership long warning	F	$t = +0.79$, wrong sign	—
flow forecasting (stop-out prediction)	B	OOS rank-corr -0.015	Law 2
factor crowding as predictor	C	dead five ways	Law 1
NMF strategy-crowding fragility	C	sign opposite thesis	Law 1
ICA reversal as tradable	E	net ≈ 0	identification, not alpha
breadth level as factor	C	net Sharpe -0.07	Law 1
imbalance reversal (bias-adjusted)	F	premium shrinks	PIT accounting bites
ML nonlinear premium	A	boosted–ridge $t = +0.27$	linear extracts it

Sample. Fifty quarters binds everything inferential: the IPCA alpha test is underpowered at this T ; the funding-beta design has two or three informative episodes; the FSS regime finding rests on a twenty-quarter window; the hyperparameter surfaces are one draw of history each.

Corporate actions. Splits are handled by the modal-atom detector; other corporate actions (mergers, spin-offs, symbol recycling) are handled conservatively by fail-closed exclusion rather than event-level treatment — a measured 0.22% contamination of exit flow, documented rather than repaired.

Book visibility. 13F books are long-only quarterly photographs: no shorts, cash, options (excluded by type), or intra-quarter turnover. The frozen-book return understates dispersion; AUM-based groupings carry leakage.

Next steps, ranked. (1) CRSP migration (coverage ceiling and delisting-inclusive shorts, with corporate-event unification via permno); (2) higher-frequency flow and performance data — monthly N-PORT portfolios and Morningstar-tier reported TNA, subscriptions/redemptions, and returns — feeding a probabilistic position *nowcast* between filings (gated on beating the frozen-book baseline at holdings reconstruction) and re-opening flow *prediction* with the instrument the literature actually uses (reported flows; implied 13F flow has no forecastable structure), with NMF-cluster hierarchical shrinkage for per-manager sensitivities; (3) accounting fundamentals and a risk model for characteristic-complete residualization and joint false-discovery control; (4) short-interest data to sharpen the SSI instrument and squeeze guard; (5) a second funding cycle to power the funding-beta design.

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A Signal definitions

B Module compendium

One block per experiment module: the question, the design choice that matters, the numbers, the verdict. Every block corresponds to a module directory with its pre-registration in the script docstring and a full report on disk.

Table 7: Quick-reference definitions (per stock, per quarter, unless noted).

signal	definition
NCB	fraction of filers opening/growing ($\geq 1.25\times$, split-adj.) a top-decile-of-book position; residualized on $\log \text{mktcap} + \log \text{ADV}$
conviction_top	fraction of filers holding the stock in their top decile
dbreadth / _common	change in number of holders (all / common filers only)
breadth_level	number of distinct holders (level; residualized)
entry/exit/net_entry	filers entering / exiting / net, over holders
dio / dio_2q	change in institutional ownership share (1q / 2q)
pso	proportion of shares outstanding held by filers
days-ADV / Δ	aggregate 13F position in days of ADV (level / change)
ti / vi	turnover / value imbalance of position changes
herf_holders	holder-concentration Herfindahl (residualized)
late_minus_early_dio	ΔIO of late filers minus early filers
pressure	$\sum_m \text{flow}_m \times \text{prior position value} / \text{ADV}$
FSS	$\sum_{m \in \text{dis}} \text{flow}_m \times \text{AUM}_m \times w_{m,i} / \text{ADV}_i$, $\text{dis} = \{\text{flow} < -0.10\}$
SSI	Tobit-inverted latent shorting propensity from the censored 13F cross-section
implied flow (manager)	$\text{AUM}_t / \text{AUM}_{t-1} - (1 + r^{\text{book}})$

Signal catalogue (general_predictive_signals). *Question:* which positioning transformations carry cross-sectional information under one protocol? *Design:* headline (raw vs. residual) fixed ex ante by the dart-throwing test; EW and VW both reported; literature anchor per signal. *Numbers:* Table 2; NCB leads at $t = +3.54$; dio/pso null as published. *Verdict:* one factor, one model ingredient, the rest documented nulls consistent with the record.

NCB implementations (champion, new_positioning). *Question:* does fresh top-decile conviction predict returns? *Design:* votes = opened-or-grown ($\geq 1.25\times$, split-adjusted) top-decile positions; residualized; full universe with zeros. *Numbers:* $t = +3.44$ to $+3.83$ across eight replications; Sharpe ≈ 1.0 – 1.1 . *Verdict:* promoted; the core factor.

Forced-sale mechanics (fire_calendar). *Question:* do distressed managers actually sell, and how? *Design:* two stages, returns only in stage two; implied flow $< -10\%$ defines distress. *Numbers:* sold fraction 34.5% vs 23.1% ($t = +10.6$; strictest contrast $+20.9$); liquidity ordering $\approx$ pro rata (-0.02). *Verdict:* mechanism validated; feeds FSS.

Stop-out exhaustion long (stopout). *Question:* do stocks bounce once a stressed owner finishes selling? *Design:* mechanism gate before price; same-crash placebo. *Numbers:* gate rate doubles (12.2% vs 6.7%, $t = +1.62$); thesis–placebo CAR_{5d} $t = +0.95$; the continuation cell is the significant one ($t = +2.09$ at 10d). *Verdict:* dead at the gate; pressure continues.

Daily stress anticipation (stress_short, v1–v3). *Question:* can drifted-book synthetic stress front-run the short? *Design:* $z(-DD) + z(\text{PainBreadth}) > 2.0$, 5-day hold, calm-manager placebo; ablations on correlation triggers and quantile cuts. *Numbers:* $t = -2.16$ (old universe) $\rightarrow t = +0.37$ (expanded, 49q). *Verdict:* **retracted** — eligibility-selection artifact; machinery survives in the monitors.

Daily monitors (daily_monitor). *Numbers:* squeeze alarm inverted-useful (alarmed shorts revert $-1.21\%/5d$, $t = -2.60$); deleveraging index conditions risk (combo Sharpe $0.49 \rightarrow 0.55$,

drawdown $-17 \rightarrow -15\%$); hot-ownership warning dead ($t = +0.79$); cover-at-exhaustion dead ($-1.5\%/q$ for 7 bps). *Verdict*: one anti-panic rule, one conditioner; no new alpha.

Flow forecasting (flow_forecast). *Question*: is implied flow forecastable one quarter ahead (Coval–Stafford construction)? *Design*: expanding coefficients; three gates; price gated on mechanism. *Numbers*: OOS rank correlation -0.015 ($t = -1.21$); predicted-distressed don’t sell ($t = +0.69$). *Verdict*: dead at gate one; Law 2.

Crash cascades (cascade). *Question*: does one position’s crash predict fund redemption and pressure on its siblings? *Numbers*: middle link $t = -1.47$; price null with clean placebo. *Verdict*: gated; the quarterly clock is too coarse for this channel.

Funding beta (flow_beta). *Numbers*: state alone $t = -0.36$ (Law 1); interaction with the common shock $t = +0.77$ on 2–3 informative episodes. *Verdict*: not yet testable; recorded, not rejected.

Window dressing (window_dressing). *Numbers*: both pre-registered signs wrong ($+1.58/-1.94$); permutation placebo follows the thesis; Q4-only reversal -1.91% (11 obs). *Verdict*: rejected; year-end anomaly footnoted.

Amendment events (restatement_shock). *Numbers*: material restatements price *against* the correction ($t = -3.52$ at +3d, clean placebo); administrative zero; reveals’ +10d drift has a contaminated placebo. *Verdict*: pre-registered thesis dead; inverted fact documented.

The clock (ragged_edge). *Numbers*: per-name entry adds $+0.21\%/yr$ long ($t = 0.18$), $+0.31\%/yr$ short ($t = -0.22$). *Verdict*: D+45 approximately optimal.

Interactions (interactions_classical). *Numbers*: five pre-registered conditional effects, none pass Bonferroni; best right-sign delta $t = +1.34$; the E4 pressure $\times$ vol flicker fell from $t = +2.06$ to $+0.77$ on fuller coverage. *Verdict*: the factor is broad; non-promotion vindicated.

Manager factor exposures (manager_factor_positioning). *Design*: $E = WX$ with placebo validation; six pre-registered hypotheses on crowding dynamics. *Numbers*: validation clean (placebo $+0.59 \rightarrow -0.16$); H3/H4/H5 dead, identical expanded. *Verdict*: measurement kept; prediction rejected.

Combiner race (score_model). *Numbers*: Section 8.1; FM wins (0.86 research protocol); NCB $\lambda t = +3.04$ and breadth-change $\lambda t = +4.01$ jointly. *Verdict*: FM is the measurement combiner; its engine decomposition is in Section 8.3.

IPCA (ipca). *Numbers*: Section 8.2; $W_\alpha p = 0.41-0.59$; unrestricted OOS Sharpe 0.97, paired $t = +2.08$ over FM; restricted ties. *Verdict*: candidate; the spanning answer stands.

Hyperparameter surfaces (champion_hyperopt, distress_hyperopt). *Numbers:* Appendix C. *Verdict:* NCB surface uniformly OOS-positive with meaningful selection; FSS surface uniform across a regime flip where in-sample selection is anti-informative.

Machine-learning cross-check (ml_positioning). *Numbers:* Section 6.4; all architectures $t = +2.6$ to $+3.4$ on the residualized target; NCB alone beats the model; orthogonalized remainder $t = +0.40$; no nonlinearity ($t = +0.27$). *Verdict:* the family’s sufficient statistic is NCB.

Smart money (wbreadth, universe_variants, patient_universe). *Numbers:* sigmoid weight effect $t = +0.06$; same-electorate baseline isolates the confound; filters hurt up to $t = -3.25$; skill persistence 0.25. *Verdict:* family closed at zero for nine.

Passive question (champion_noetf, champion_nopassive, follow_nopassive, breadth_universes). *Numbers:* Section 3; ranking correlation 0.974 after ETF purge; passive removal +6 bps. *Verdict:* elephants without votes.

Copycat graph (copycat, copycat_pop). *Numbers:* graph stability 0.94; follower hangover $t = -2.5$; popularity window closed. *Verdict:* instrument kept, strategy rejected.

Synthetic short interest (ssi). *Numbers:* Tobit inversion; rank correlation with actual short interest 0.55 (from 0.47 pre-fix). *Verdict:* instrument validated; strengthened by better data.

Latent structure (ica_demand, original_methods). *Numbers:* Section 9; rotation placebo $+1.29$ vs -0.77 ; stability 81%; NMF conditional fragility sign opposite to thesis. *Verdict:* identification instruments; alpha modest and labeled.

Crowding graveyard (grafo_crowding, absorcao, overhang). *Numbers:* diffusion thesis wrong sign ($+1.51$); a flow-vulnerability centrality variant added nothing over the plain size proxy and was dropped; absorption and overhang refinements $t = +0.98 / -0.05$. *Verdict:* more entries under Law 1.

Fund performance (fund_performance). *Numbers:* Section 11; median -3.3% /yr before fees; $IR > 0$ 13%; persistence $+0.25$. *Verdict:* the descriptive base rate behind the smart-money law.

C Hyperparameter surfaces

The bootstrap selection protocol, self-contained. Following Oliveira, Guzman and Firoozye [2025]: (1) take each configuration’s in-sample *quarterly* return series; (2) moving-block bootstrap — blocks of four consecutive quarters, preserving short-range autocorrelation and volatility clustering, $B = 2,000$ resamples; (3) *coupled* indices: the same resample paths for every configuration, so cross-config differences are configuration effects, not resampling luck; (4) compute the utility (Sharpe, or IR for the long-only product) on each resample, giving one distribution per configuration; (5) select on the 5th percentile — a configuration wins only by being good in its bad resamples, which penalizes performance concentrated in a few lucky quarters (the winner’s-curse correction that a point-estimate maximum lacks). The choice is then frozen and evaluated on untouched years. Scope

and limits, stated plainly: the bootstrap is a robust *ranking* criterion, never a source of formal inference (all p -values in this report are Newey–West); block length (one year) and the percentile were fixed a priori; at $T \approx 30$ the percentile tracks the point estimate (reported as an honest null); and the protocol assumes a stationary utility distribution — an assumption it can itself audit through the cross-configuration in-sample to held-out rank correlation, which validated selection once (NCB, +0.53) and refused it twice (FSS, LO tilt).

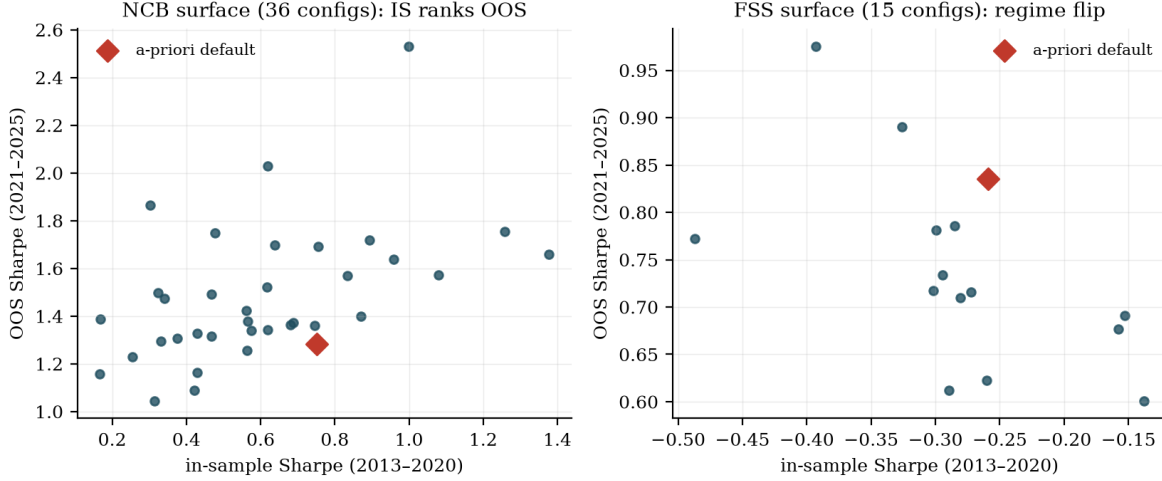


Figure 12: In-sample (2013–2020) versus out-of-sample (2021–2025) Sharpe by configuration. Left: NCB, 36 configurations — the whole surface is OOS-positive and the in-sample ordering predicts the OOS ordering (rank corr. +0.53); selection is meaningful, the default (diamond) is conservative mid-surface. Right: FSS, 15 configurations — the surface moves as one block across a regime flip and the IS ordering *anti-predicts* OOS (rank corr. −0.81): the correct use of the selection protocol is to refuse to select.

Table 8: NCB surface, top rows and default (full 36-row table in the repository). Coupled moving-block bootstrap (block 4, $B = 2,000$); selection statistic = 5th percentile of bootstrap Sharpe.

config (w-quantile / growth / min holders)	IS Sharpe	boot p05	OOS Sharpe	OOS t
0.95 / 1.50 / 3 (both selectors' pick)	+1.38	+0.92	+1.66	+4.85
0.95 / 1.50 / 5	+1.26	+0.72	+1.76	+5.08
0.95 / 1.50 / 10 (best OOS)	+1.00	+0.49	+2.53	+5.44
0.90 / 1.25 / 5 (a-priori default)	+0.75	+0.14	+1.28	+3.31
worst on grid	+0.17	−0.34	+1.39	+3.93

Table 9: FSS surface (threshold $\times$ basket fraction), index-hedged short-leg Sharpe. All fifteen configurations lose modestly in-sample and pay out-of-sample: a regime, not a hyperparameter.

config	IS Sharpe	boot p05	OOS Sharpe	OOS t
−0.05 / top 10%	−0.14	−0.57	+0.60	+1.87
−0.10 / top 20% (default)	−0.26	−0.75	+0.84	+2.84
−0.20 / top 20% (best OOS)	−0.39	−1.05	+0.98	+4.02